

E-Commerce Sales Analysis

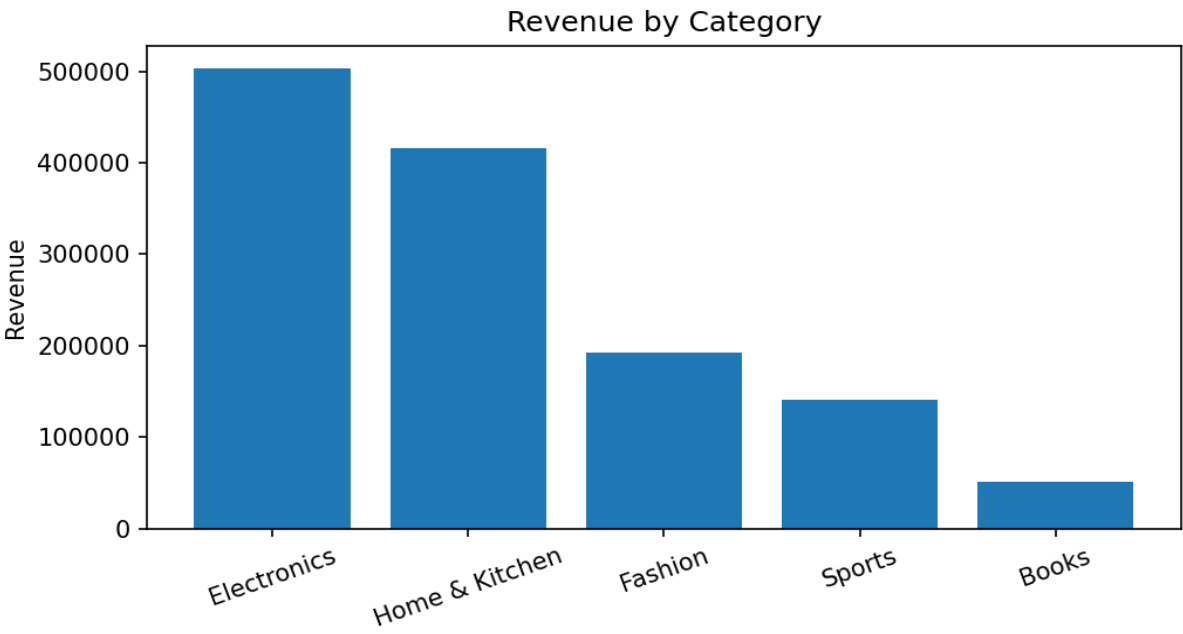
Complete Jupyter Notebook Project — Output Report

A complete Python analytics project covering data validation, KPIs, category performance, monthly trends, regional analysis, top products, payment methods, discounts, correlation and recommendations.

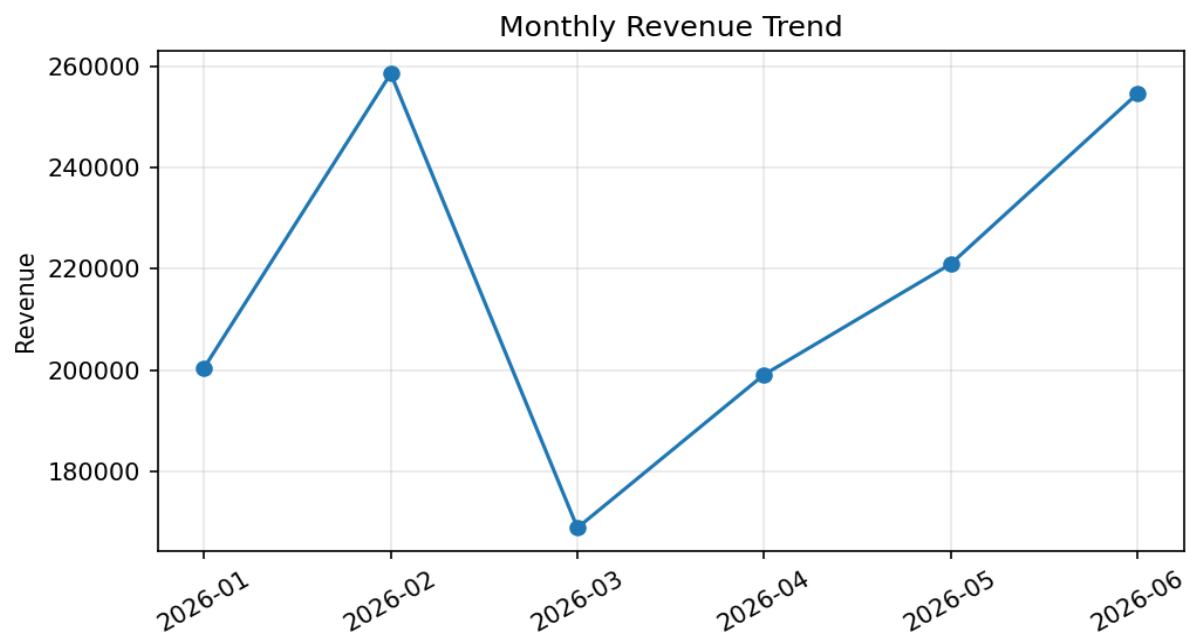
Metric	Value
Total Revenue	■1,302,689.30
Total Profit	■437,574.04
Units Sold	534
Average Order Value	■7,237.16
Profit Margin	33.59%

Category Performance

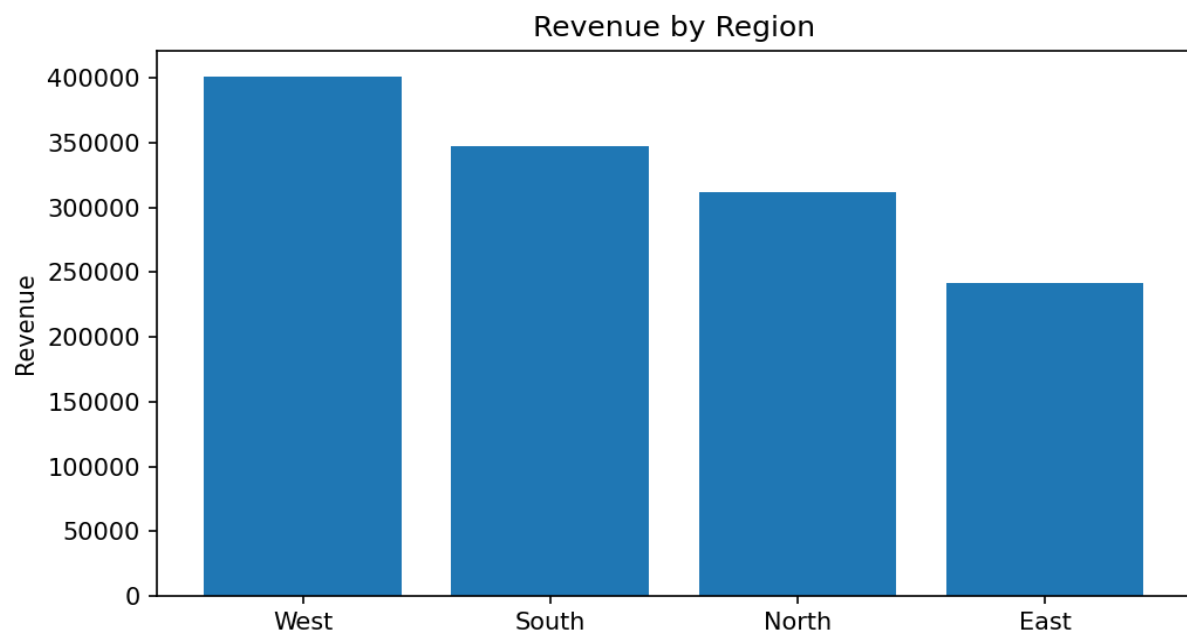
Category	Revenue	Profit	Units	Margin
Electronics	■502,834	■177,354	129.0	35.27%
Home & Kitchen	■416,070	■129,639	131.0	31.16%
Fashion	■191,813	■65,917	120.0	34.37%
Sports	■141,334	■47,596	83.0	33.68%
Books	■50,639	■17,068	71.0	33.71%



Monthly Revenue Trend



Regional Revenue



Top Products

Product	Revenue	Units
Bluetooth Speaker	■156,675	40.0
Power Bank	■132,999	28.0
Cookware Set	■123,165	41.0
Air Fryer	■118,800	38.0
Wireless Headphones	■118,302	40.0
Mixer Grinder	■103,376	29.0
Smart Watch	■94,858	21.0
Coffee Maker	■70,728	23.0
Running Shoes	■65,130	37.0
Backpack	■57,194	35.0

Key Findings

- Highest revenue category: Electronics
- Highest revenue region: West
- Highest revenue product: Bluetooth Speaker
- Highest revenue month: 2026-02
- Highest revenue payment method: Cash on Delivery
- Overall profit margin: 33.59%

Recommendations

- Focus inventory planning on high-revenue categories.
- Strengthen regional campaigns and investigate weaker regions.
- Promote top products through bundles and cross-selling.
- Monitor discount levels and their effect on profit margins.
- Support popular payment methods.
- Use monthly trends for inventory and promotional planning.

The Jupyter Notebook contains the complete Python workflow. The CSV is included so the notebook can be run independently.